Charges and Charging, and Coulomb's Law

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1 Charging Introduction

These are my notes on charges and charging for Physics!

Charges are fundamental properties for particles, just like **masses** and **spins**.

2 Fundamental Properties of Particles

There are three particles that matter for charges and charging.

Particle	Charge (in e)	Charge (in Coulombs (C))
Proton	$1e$	1.6×10^{-19} C
Electron	- $1e$	-1.6×10^{-19} C
Neutron	$1e$	1.6×10^{-19} C

b
Charges are usually sent in multiples of elementary charge, usually through electrons (because if you remove protons, the atoms themselves change).

Note: Elementary charges are elementary because they are the smallest stable charge. You can get quarks (with fractions of elementary charges), but will reform to these particles.

3 Sending and Receiving Charge

There are three methods for sending and recieving charge:

- Friction
- Conduction
- Induction

3.1 Friction

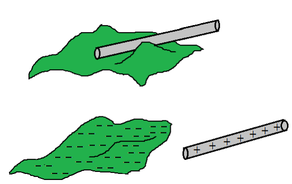


Figure 1: Example of Friction

When two objects rub with each other, the object with higher affinity will **take** electrons from the other.
Usually, if you have two of the same object rub the same way, the objects have the same charge.

3.2 Conduction

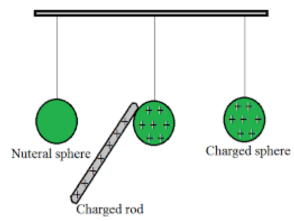


Figure 2: Example of Conduction

When two conductors (all objects are conductors in the end, if you try hard enough). If objects are identifical, their charge will even out. Elsewise, they will just equalize. If I remember correctly, applying the "same" friction will cause the same charge result.

Note: These conductors equalize to minimize the system's *kinetic energy*.

3.3 Induction

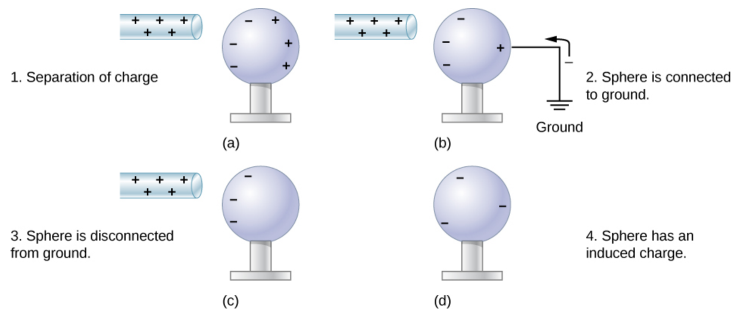


Figure 3: Example of Induction

A charge can change through proximity to an object. To equalize the charge afterwards, you can just tap the two objects together (if I remember correctly).

3.4 Note: Polarized Neutral

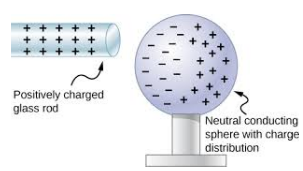


Figure 4: Example of a polarized neutral

A neutral object can be polarized through having the charged object attract the opposite charge of a neutral object, causing attraction through the distance.

3.5 Note: Question Stuff

The question worksheet was *Electric Charge and Methods of Charging*, and the only weird thing was question 4. To solve, just do

$$\frac{1.5\,\mathrm{m} \times \left(6.023 \times 10^{23}\,\text{particles per m}\right)}{1 \times 10^{6}\,\text{particles per e}} \times \frac{1.6 \times 10^{-19}\,\mathrm{C}}{e} = 0.144\,\mathrm{C}$$

6.023×10^{23} is Avogadro's (Avocado's) number, which is equal to the number of particles per mole. One just has to divide by that the number of particles before one is charged, and then to convert elementary charges to coulombs.

4 Coulomb's Law

Coulomb's Law is related to how two charged particles are attacted to each other.

4.1 Coulombs's Law from Regents

$$F_e = \frac{kq_1q_2}{R^2}$$

Problems with this formula:

- k assumes a vacuum
- q_1 , q_2 are point particles
- One has to assume one object is stationary

4.2 Building from Coulomb's Law for Physics C

1. We don't want to use k anymore. Instead we use $\frac{1}{4\pi\epsilon}$. ϵ represents the permittivity of a material [1]. For vacuums, the permittivity, or ϵ_0, is equal to $8.85 \times 10^{-12} \frac{\mathrm{C}^2}{\mathrm{Nm}^2}$
2. To fix the fact that that q_1 and q_2 must be point particles, one will have to integrate over δq, or change in charge, to find the total charge, and integrate over δF. to find the force.

Putting everything together, we have the new formula:

$$\boxed{\vec{F}_e = \frac{1}{4\pi\epsilon}\frac{q_1q_2}{R^2}\hat{r}}$$

where $\hat{r}$ is vector pointing to the other charge, making it so that when the equation positive, the resultant vector is pointed away from the other particle. Remember, $\boxed{\epsilon_0 = 8.85 \times 10^{-12}\frac{\mathrm{C}^2}{\mathrm{Nm}^2}}$ [this is also on the reference table]

5 Electrical Fields

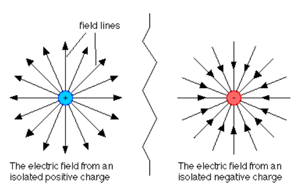


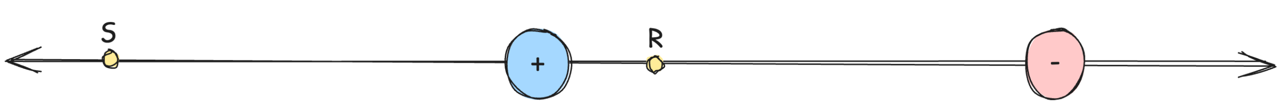
Figure 5: Examples of a Electrical Field

The formula for calculating a electirc field, which is the force on the probe charge divided by the charge:

$$\vec{E}(r) = \frac{\vec{F}_0(r, q_0)}{q_0} = \frac{\frac{1}{4\pi\epsilon_0}\frac{Qq_0}{r^2}}{q_0} = \boxed{\frac{1}{4\pi\epsilon_0}\frac{Q}{r^2}\hat{r}\frac{\mathrm{N}}{\mathrm{C}}}$$

NOTE: Always assume that the satellite particle is positive.

Example
The figure below shows a proton p and an electron e on an x-axis. What is the direction of field due to the electron at points S and R, and then the net charge.



[1]the ability of a particle to hold energy at a specific field without breaking down

- The electron's field on S is towards the electron in a rightward direction, and at R is towards the electron in a left direction
- The net field at S is a lesser (lesser magnitude) arrow towards the right, and at R its a larger arrow (greater magnitude) to the left.

5.1 Calculating the Electrical Charge of Non-Point Particles

If we have a long thin rod, we have a few methods to try to integrate and find the field:

- You can integrate (**DRAWBACKS**: Both wierd directionally and changes with the charge too, not always ideal)
- Chickening out and using a ∞ lengthed rod, or use **Gauss' Law** (ohhhhh...)

6 Calculating Continious Charge Distributions

Reminder: It's arbitrarily difficult to calculate the charge of a continious shape. Therefore in order to solve these problems, one has to simplify stuff.
Simplification 1: The field exerted in point R by a charged ring.

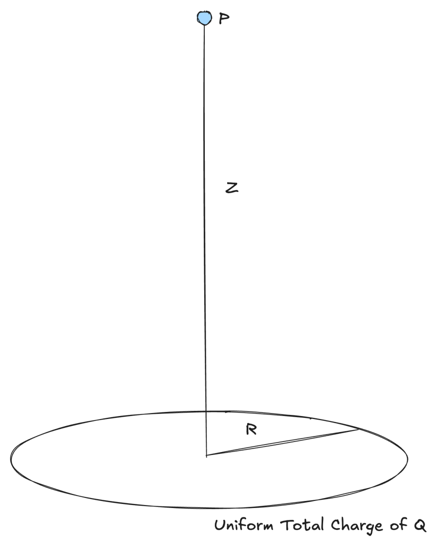


Figure 6: the Ring in question

Simplications (it's a continious distance, but):

- Symmetry
- Rigid Structure

6.1 Deriving the Formula for Calculating the Electrical field

Let's divide this method into several steps

1. Define Charge Density
2. Use symmetry
3. Integration

***Step 1*: Defining Charge Density**
Let's create a new value, and let's define it as charge density, λ, to help make this easier.

$$\lambda = \frac{\delta q}{\delta l}$$

Also, from the special case of it being a ring, we know that charge density is also the total charge minus the radius of the ring:

$$\text{Charge Density} = \frac{Q}{2\pi r}$$

Then, we know that:

$$\lambda = \frac{\delta q}{\delta l} = \frac{Q}{2\pi r}$$
$$\delta q = \lambda \delta l$$

***Step 2*: Finding $\delta\vec{E}$ from δq**

Let's first start with integrating and then substituting for the change in charge.

$$\delta\vec{E} = \frac{1}{4\pi\epsilon R^2}\delta Q = \frac{1}{4\pi\epsilon}\frac{\lambda\delta l}{z^2+r^2}$$

The visualization of the new figure:

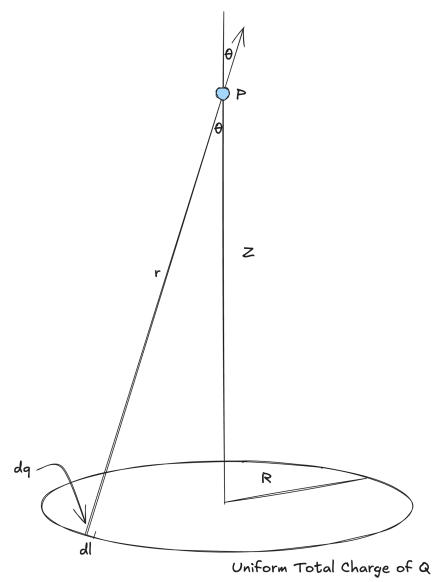


Figure 7: the Ring in question

***Step 3*: Use Symmetry**

Common sense time: if you have two little $\delta\vec{F}$ from two little δq from two δl which are on opposite sides of circle, they cancel out. This is where we get $\delta\vec{E}_{\text{VERT}}$. After we integrate that, we should do some substituting

$$\delta E_{\text{VERT}} = \left|\delta\vec{E}\right|\cos\theta = \frac{1}{4\pi\epsilon}\frac{hdl}{(z^2+d^2)}\cos\theta$$
$$= \frac{1}{4\pi\epsilon}\frac{hdl}{(z^2+d^2)} \times \frac{z}{r}\left[\cos\theta = \frac{z}{R}\right]$$
$$= \frac{1}{4\pi\epsilon}\frac{\lambda\delta l}{(z^2+R^2)^{3/2}}z$$

What's so great about this is that everything is constant, so integrating over the length of the ring would be trivial.

***Step 4*: Integrate**

$$\vec{E} = \int \delta E_{\text{VERT}} = \frac{1}{4\pi\epsilon}\frac{\lambda z}{(z^2+R^2)^{3/2}}\int_0^{2\pi R}\delta l$$
$$= \frac{\frac{Q}{2\pi R}z}{4\pi\epsilon(z^2+R^2)}2\pi R\left[\textbf{NOTE}: \lambda = \frac{Q}{2\pi R}\right]$$
$$= \boxed{\frac{Qz}{4\pi\epsilon(z^2+R^2)^{3/2}}}$$

Some Test Cases

- $E_{z=0} = 0$
- $E_{z>>R} = \frac{Q}{4\pi\epsilon R^2}$ [basically the point formula, its very far away it behaves like a point particle]